

ing computer graphics have traditionally been described by application developers as “embarrassingly parallelisable”. What that means is, the graphics processing task can be broken down into many stages and components. The interesting thing about both the ATi and nVidia offerings, is the way each handles parallelism in graphics processing.

The Radeon X800 XT sports 16-pixel-wide pipelines, split into groups of four—with four ‘regular’ pipelines beneath each ‘wide’ pipeline. Each wide pipeline works with groups of four pixels, making for a total of 16 pixels. Each regular pipeline has its own pixel shader and texture unit, as well as shared cache memory across the wide pipeline.

The quad pipelines can be enabled or disabled; thus, the X800 can have from eight to 16 pipelines in total. Lower-end models have eight, and higher end

sors still remains at 24 bits per colour channel—red, blue, green and alpha—the same as the 9800 XT. The data formats supported are both 16-bit and 32-bit. However, the computational power is significantly higher, with twice as many arithmetic/logic units (ALUs). With this, the X800 can process up to five instructions for each pixel per clock cycle, as opposed to three with the R3x0 family. The X800 also scores from its predecessor with a vast increase in the number of mathematical and texture instructions. This means that the X800 XT can perform complex pixel shader calculations more efficiently. If the complex pixel-shading processing cannot be carried out in a single pass, the job is split, and each fraction is processed. The results are stored in temporary buffers and aggregated later.

The GeForce 6800 Ultra also has a 16-pixel pipeline unit—a big jump compared to the GeForce FX5950’s four-pipeline design. Each 16-pixel pipeline of the NV40 comes with its own texture unit per pipeline. This differs from the GeForce FX5950’s 4-pixel pipeline—two texture units per pipeline, which could output up to 8 pixels per clock cycle. The 6800, on the

The games that people play

For the sort of hard cash you will shell out for owning one of the top graphics cards, you definitely need to have the games too. Both cards are DirectX 9.0 compliant, so the games you could start with, and drool over are *Far Cry*, and *UT 2004*, as well as *Prince of Persia: The Sands of Time*, *Splinter Cell: Pandora Tomorrow* and *PainKiller*, which are already out on the store shelves. There are more games coming in the pipeline, though. Look forward to playing *S.T.A.L.K.E.R.: Shadow of Chernobyl* and *Half-Life 2* as well. These games aren’t tailor-made for either of the cards, but you can enjoy all of these games the way the developers intended, full with all the special effects and flashes, bangs and pops and gore. id’s *DOOM 3* would also be a worthy game for them, as and when it’s released.

ways too. First, the number of temporary registers has been increased—this used to be a weak point in the FX 5950 series. The pixel processors can now perform floating point computations with full 32-bit precision per colour channel, and with the additional temporary registers.

With changes in the internal programmability of the GPUs, we can expect to see additional precision in computing shaders. Developers can now employ more advanced rendering techniques with fewer compromises.

Both ATi and nVidia have reworked and vastly enhanced the vertex processing capabilities of their offerings, and the result of this will be seen while rendering complex game models. With a huge amount of geometric data to be processed, for the many polygons in each frame, the vertex processing capability of the nVidia and ATi cards will be welcome. ATi’s enhanced instruction set will allow for more efficient pixel and texture computations, but the new Radeon does not support the pixel and vertex shader 3.0 specifications of DirectX

The GRAPHIC Details

After more than two years of rumours and ‘informed guesses’, nVidia and ATi have come out with their latest graphics blockbusters. Find out what’s changed

models—such as the X800 XT Platinum Edition—have sixteen. The X800 now comes with six vertex shader pipelines—two more than the 9800 XT. ATi claims that this, and the increased GPU clock speeds, gives the X800 twice the vertex shading power of the 9800 XT.

The X800 now also has more temporary registers, which allow for more efficient handling of complex shading operations. The internal floating point calculation precision in the X800’s pixel proces-

other hand, manages 16 pixels within a clock cycle—the same as the Radeon X800. For today’s games, with large and complex models, and many polygons in each scene, the 6800’s six vertex shaders provide dazzling vertex processing performance. Think of realistic, high resolution gameplay, with fascinating details—that’s the result of just the sort of serious number crunching that works here.

nVidia has enhanced the computational abilities of the NV40 family in other

Info

The nVidia GeForce 5950 FX is based on the the NV38 GPU architecture, and the nVidia GeForce 5700 FX is based on the NV36 GPU.

The GeForce 6800 Ultra is the first example of the the NV40 GPU.